Signals & Systems **Test-I** **Date:19.9.2024**

Attempt all questions **M.M-20**

1. Determine if the given discrete time signal $x[n] = 2\cos\left(\frac{\pi}{4}n\right) + \sin\left(\frac{\pi}{8}n\right) - 2\cos\left(\frac{\pi}{2}n + \frac{\pi}{6}\right)$ is periodic. If yes, determine its fundamental period. 2
2. Determine whether the given system $y(t) = x(t/3)$ may or may not be memoryless, time-invariant, linear, causal. 4
3. Find the response of the discrete time system with unit impulse response $h(n) = \left[2 \quad \underset{\uparrow}{1} \quad 3\right]$ to the input $x(n) = \left[\underset{\uparrow}{1} \quad 2 \quad 3\right]$. 3
4. For the given signal $x(t)$ in Fig. 1(a), Sketch and label the following signals $x(t-1), x(-t+2), x(2t), x(t/2)$. 4
5. The input and the impulse response of a system are given by $x(t) = u(t+2)$ and $h(t) = u(t-3)$, respectively. Determine the system output using graphical method. 5
6. Express $x(t)$ given in Fig. 1(b) in terms of $g(t) = u(t+1) - u(t-1)$ 2

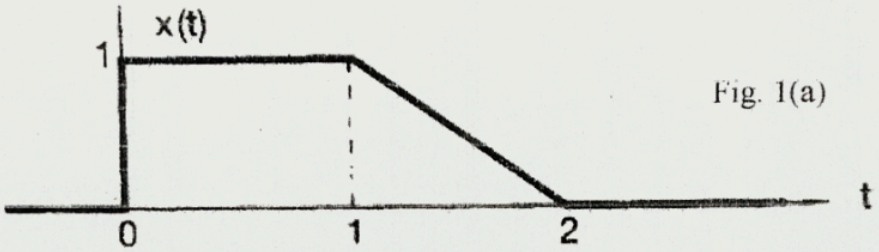

Fig. 1(a)

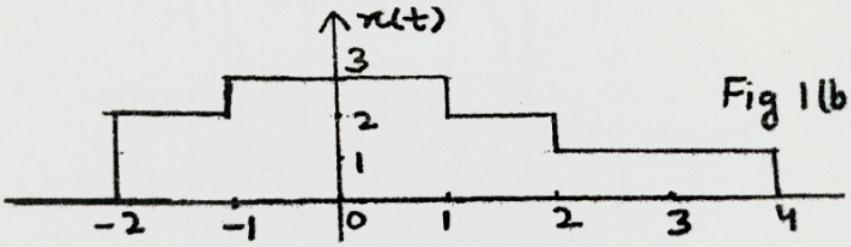

Fig 1(b)